

UNIX - PIPES AND FILTERS

<http://www.tutorialspoint.com/unix/unix-pipes-filters.htm>

Copyright © tutorialspoint.com

You can connect two commands together so that the output from one program becomes the input of the next program. Two or more commands connected in this way form a pipe.

To make a pipe, put a vertical bar | on the command line between two commands.

When a program takes its input from another program, performs some operation on that input, and writes the result to the standard output, it is referred to as a *filter*.

The grep Command

The grep program searches a file or files for lines that have a certain pattern. The syntax is –

```
$grep pattern file(s)
```

The name "grep" derives from the ed *aUNIXlineeditor* command g/re/p which means "globally search for a regular expression and print all lines containing it."

A regular expression is either some plain text *aword, forexample* and/or special characters used for pattern matching.

The simplest use of grep is to look for a pattern consisting of a single word. It can be used in a pipe so that only those lines of the input files containing a given string are sent to the standard output. If you don't give grep a filename to read, it reads its standard input; that's the way all filter programs work –

```
$ls -l | grep "Aug"
-rw-rw-rw-  1 john  doc      11008 Aug  6 14:10 ch02
-rw-rw-rw-  1 john  doc      8515 Aug  6 15:30 ch07
-rw-rw-r--  1 john  doc      2488 Aug 15 10:51 intro
-rw-rw-r--  1 carol doc      1605 Aug 23 07:35 macros
$
```

There are various options which you can use along with grep command –

Option	Description
--------	-------------

- | | |
|----|---|
| -v | Print all lines that do not match pattern. |
| -n | Print the matched line and its line number. |
| -l | Print only the names of files with matching lines <i>letter " l "</i> |
| -c | Print only the count of matching lines. |
| -i | Match either upper- or lowercase. |

Next, let's use a regular expression that tells grep to find lines with "carol", followed by zero or more other characters abbreviated in a regular expression as ".*"), then followed by "Aug".

Here we are using -i option to have case insensitive search –

```
$ls -l | grep -i "carol.*aug"
-rw-rw-r--  1 carol doc      1605 Aug 23 07:35 macros
$
```

The sort Command

The **sort** command arranges lines of text alphabetically or numerically. The example below sorts the lines in the food file –

```
$sort food
Afghani Cuisine
Bangkok Wok
Big Apple Deli
Isle of Java
Mandalay
Sushi and Sashimi
Sweet Tooth
Tio Pepe's Peppers
$
```

The **sort** command arranges lines of text alphabetically by default. There are many options that control the sorting –

Option	Description
-n	Sort numerically <i>example: 10willsortafter2</i> , ignore blanks and tabs.
-r	Reverse the order of sort.
-f	Sort upper- and lowercase together.
+x	Ignore first x fields when sorting.

More than two commands may be linked up into a pipe. Taking a previous pipe example using **grep**, we can further sort the files modified in August by order of size.

The following pipe consists of the commands **ls**, **grep**, and **sort** –

```
$ls -l | grep "Aug" | sort +4n
-rw-rw-r-- 1 carol doc      1605 Aug 23 07:35 macros
-rw-rw-r-- 1 john  doc      2488 Aug 15 10:51 intro
-rw-rw-rw- 1 john  doc      8515 Aug  6 15:30 ch07
-rw-rw-rw- 1 john  doc     11008 Aug  6 14:10 ch02
$
```

This pipe sorts all files in your directory modified in August by order of size, and prints them to the terminal screen. The sort option `+4n` skips four fields *fieldsareseparatedbyblanks* then sorts the lines in numeric order.

The pg and more Commands

A long output would normally zip by you on the screen, but if you run text through **more** or **pg** as a filter, the display stops after each screenful of text.

Let's assume that you have a long directory listing. To make it easier to read the sorted listing, pipe the output through **more** as follows –

```
$ls -l | grep "Aug" | sort +4n | more
-rw-rw-r-- 1 carol doc      1605 Aug 23 07:35 macros
-rw-rw-r-- 1 john  doc      2488 Aug 15 10:51 intro
-rw-rw-rw- 1 john  doc      8515 Aug  6 15:30 ch07
-rw-rw-r-- 1 john  doc     14827 Aug  9 12:40 ch03
.
.
.
-rw-rw-rw- 1 john  doc     16867 Aug  6 15:56 ch05
--More--(74%)
```

The screen will fill up with one screenful of text consisting of lines sorted by order of file size. At the

bottom of the screen is the **more** prompt where you can type a command to move through the sorted text.

When you're done with this screen, you can use any of the commands listed in the discussion of the `more` program

Loading [MathJax]/jax/output/HTML-CSS/jax.js